

Chirag Deepak Shinde

DATA ENGINEER

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SUMMARY

Results-driven Data Engineer with 3+ years building streaming and batch pipelines across banking and healthcare: real-time fraud and AML detection, HL7-to-FHIR interoperability, and production-grade solutions strengthening compliance and decision-making.

Skilled in designing distributed ETL and ELT pipelines and end-to-end workflows with Python, SQL, Apache Spark, Kafka, and Airflow across AWS and Azure, transforming raw data into clean, high-quality datasets that drive reliable enterprise analytics and reporting.

Experienced integrating machine learning and LLM-powered solutions like RAG and vector search into pipelines, focused on automating insight extraction, reducing manual effort, and delivering business value across regulated banking and healthcare environments.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, Scala, Java, Bash
- **Data Engineering:** ETL/ELT pipeline design, Apache Spark, dbt, data warehousing, data modeling, data quality frameworks
- **Orchestration & Streaming:** Apache Airflow, Apache Kafka, Spark Structured Streaming, batch & real-time processing
- **Cloud Platforms:** AWS (S3, Lambda, EMR, Glue, SageMaker), Azure (Data Factory, Synapse, Azure ML)
- **Databases & Warehouses:** PostgreSQL, MySQL, MongoDB, Snowflake, Redis, Pinecone / vector DBs
- **Platforms & MLOps:** Databricks, Docker, Kubernetes, MLflow, Terraform, CI/CD, model deployment & monitoring
- **LLM / Generative AI:** RAG pipelines, LangChain, embeddings, vector search, prompt engineering, LLM evaluation
- **Version Control & Tools:** Git, GitHub/GitLab, Jira, Linux

EXPERIENCE

Bank of America | *Data Engineer*

Sep 2025 - Present

- Developed a scalable transaction streaming pipeline supporting fraud and AML monitoring, processing 5M+ daily transactions across multiple business lines, enabling near-instant suspicious-activity detection and real-time analytics for enterprise risk and compliance teams.
- Built fault-tolerant, distributed Apache Kafka ingestion for high-throughput transaction events, persisting raw data to Amazon S3 as a centralized data lake so batch backfills and streaming jobs read one shared source, ensuring reliable flow into fraud-monitoring systems.
- Engineered robust, modular ETL and ELT pipeline design and data modeling with SQL and Python, structuring raw transaction, customer, and account records into clean, standardized datasets that strengthened accuracy of downstream fraud and AML detection logic.
- Moved fraud scoring from batch to Spark Structured Streaming, integrating Redis for low-latency real-time feature lookups, reducing fraud-alert latency from roughly 15 minutes to near real time for risk and compliance teams through scalable, distributed stream processing.
- Orchestrated distributed data-processing jobs with Apache Airflow on AWS, using AWS Lambda for event-driven triggers, delivering, cost-optimized workflows that lowered operational costs by 25% through improved pipeline scheduling across the fraud platform.
- Automated data-quality checks (schema, null, and range validation) within CI/CD pipelines, and tracked model versions, parameters, and metrics using MLflow, enabling reliable deployment and reproducible with end-to-end observability across the fraud platform.
- Built the data pipeline feeding an LLM alert-summarization tool, preparing and indexing case data for retrieval-augmented generation, adopted by fraud and compliance teams to speed up fraud-alert triage, strengthen AML governance, and improve operational efficiency.

Accenture | *Data Engineer*

Jun 2022 - Jul 2024

- Implemented an end-to-end healthcare data ingestion and FHIR interoperability pipeline, reducing manual data-integration effort by 45% while enabling standardized, compliant patient-record exchange across multiple clinical systems for a large US healthcare client.
- Delivered configurable, Azure Data Factory pipelines ingesting scheduled batch workloads of HL7 v2 clinical messages, mapping patient, encounter, and lab data from multiple hospital EHR and laboratory systems into standardized, schema-validated downstream datasets.
- Created ETL and ELT transformation logic with Python and SQL, mapping and validating raw HL7 data into FHIR R4 resources through structured data modeling that standardized patient, condition, and observation records for downstream clinical analytics teams.
- Utilized Apache Spark on Databricks for distributed processing, cleansing, and deduplication of clinical datasets, accelerating curated data delivery by 35% while ensuring accurate, high-quality FHIR datasets for healthcare interoperability and reporting workflows.
- Supported automated pipeline orchestration using Apache Airflow to schedule, monitor, and manage dependencies across ingestion and transformation jobs, loading curated FHIR datasets into Azure Synapse to enable consistent data access for clinical reporting teams.
- Designed a RAG pipeline with embeddings and semantic vector search, applying prompt engineering, model evaluation, and LLM evaluation that improved FHIR field-extraction accuracy by 30% over the prior keyword-based method from unstructured clinical notes.
- Implemented PHI de-identification using HIPAA Safe Harbor and validation controls, producing governed, interoperable FHIR datasets adopted by analytics and compliance teams, reducing data-compliance review effort by 40% across the enterprise healthcare data platform.

EDUCATION

University of Illinois at Chicago | *Master of Science, Computer Science*

Chicago, USA Aug 2024 - May 2026

Savitribai Phule Pune University | *Bachelor of Engineering, Computer Engineering*

Pune, India Aug 2019 - May 2023

CERTIFICATIONS

• Databricks Certified Data Engineer Associate: Databricks

Jun 2026